

Project Overview

This project aims to modernize a small library by creating a cohesive database system for the inventory and memberships. The database will store all items in the library, track transactions by clients, generate reports, generate fines, and maintain a record of other relevant information.

Scope

On the staff side, the system should be able to handle storage of a variety of media with associated information, track borrowing and return transactions, and also generate reports of trends, fines, and client activity. On the client side, the system will track ID, name, and contact information, and enforce membership fees/limits/etc.

Glossary

1. SSN - Social Security Number
2. ISBN - International Standard Book Number
3. ISSN - International Standard Serial Number
4. DOB - Date of Birth

Entities

Staff

- **Staff ID (int)**
- Name (String)
- Age (int)
- Phone Number (string)
- Address (string)
- SSN (int)
- DOB (str)

Client

- **Client ID (int)**
- Name (str)
- Contact Information (str)
- Membership (str)
- Account Status (str)

Copy

- **Copy ID (int)**
- Type (book, digital media, magazine, etc)
- Availability Status (bool)

Book

- **ISBN (int)**
- Title (str)
- Author (str)
- Publication Year (int)
- Genre (str)
- Count (int)

Digital Media

- **ISBN (int)**
- Title(str)
- Format
- Creator(str)
- Release Year
- Genre(str)
- Count (int)

Magazine

- **ID (int)**
- Title(str)
- Issue Number (int)
- Publication Date (str)
- Count (int)

Transaction

- **Transaction ID (int)**
- Checkout Date (str)
- Due Date (str)
- Return Date (str)
- Copy ID (int)
- Copy Type (str)

Reports

- **Report ID (int)**
- Date of Generation (str)
- Type (Fee, trends, overdue, revenue, summary) (str)

Fine

- **Fine ID (int)**
- Copy ID (int)
- Client ID (int)
- Due Date (str)
- Fine Amount (float)
- Date Paid (str)

Membership

- **MembershipTypeID (int)**
- Typename (str)
- BorrowLimit (int)
- DailyFineRate (float)

Reservation

- **Reservation ID**
- ClientID (int)
- ItemID (int)
- Reservation Date (int)
- Status (str)

Relationships

Staff check out Copy

Staff return Copy

Staff add new Copy

Staff manage Clients

Staff generate Reports

Clients reserve Copy

Clients return Copy

Clients create Transaction

Clients makes Reservation

Clients pay Fine

Clients have Membership

Transaction generates Fine

Transcation checks out Copy

Reservation reserves Copy